

# Anjaneya Bhardwaj

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## INTRODUCTION

As an aspiring Software Engineer currently pursuing an MS in Artificial Intelligence at the University at Buffalo, I focus on building robust, scalable data platforms and high-performance AI services. My experience includes designing healthcare data systems, optimized RESTful APIs, containerized workloads and using AI tools like copilot to solve real world complex problems. I am committed to leveraging my expertise in data-driven solutions to empower organizations.

## EDUCATION

### University at Buffalo, SUNY

2024 - May 2026

*MS, Artificial Intelligence*

- Developed expertise in Generative Diffusion Models, Reinforcement Learning, and Neural Network design. Led collaborative teams to build and optimize predictive models, managing high-performance data workflows and scaling neural architectures.
- Coursework:** Data Intensive Computing, Numerical Methods for Data Science, Reinforcement Learning, Computer Vision, Pattern Recognition, Basics of AI, Intro to ML

### Jaypee Institute of Information Technology (JIIT)

2020 - 2024

*B.Tech, Information Technology*

- Developed skills in problem solving, leadership, time management, product planning, solution delivery, and cross-team collaboration over four years of engineering.
- Coursework:** Data Mining, Web Algorithms, Information Retrieval, Machine Learning, Big Data, Financial Accounting

## EXPERIENCE

### ECHO India

Jun 2023 - Jul 2023

*Data Science Intern*

- Collaborated with senior data scientists and analysts to collect, clean, and preprocess datasets from diverse sources using Python and pandas, enabling downstream predictive modeling
- Created visualizations to illustrate key trends in user behavior, helping stakeholders quickly identify areas for improvement
- Performed **NLP** tasks and built a sentiment analyzer with the **VADER** lexicon in Python, automatically classifying user feedback and highlighting positive and negative trends for the product team

## CERTIFICATIONS

- Supervised Machine Learning- Regression and Classification: Issued Nov 2024.** Learned about regression for predictive numerical analysis and classification for accurate data categorization for robust decision-making in diverse applications. Online Course by Stanford, DeepLearning.AI.
- Affective Computing: Issued May 2024.** Gained skills in emotion recognition, computational modeling of emotions, and analysis of multimodal data (voice, facial expressions, physiological signals) using machine learning and signal processing techniques. Developed an understanding of ethical, legal, and social implications in human-machine interaction. Online Course by NPTEL, IIIT-D.

## PROJECTS

### Healthcare Data Management and Warehousing System | [GitHub](#)

- Designed a normalized PostgreSQL schema (3NF) and a Star Schema data warehouse optimized for complex analytical queries and high-performance reporting.
- Built an automated pipeline using Python and **dbt** to engineer Fact/Dimension tables for hospital revenue and patient workload metrics.
- Containerized** the full stack (Database, API, and Pipeline) using **Docker and Docker Compose**, ensuring consistent development and deployment across environments.
- Implemented performance tuning and database indexing, reducing execution time from **616ms to 53ms** (approx. 90% improvement)

### Multi-Agent RL for Network Intrusion Detection | [GitHub](#)

- Developed an AI-driven cybersecurity defense system utilizing **Multi-Agent Reinforcement Learning (MARL)** to protect critical network assets under resource constraints.
- Modeled enterprise networks as graphs using **GNNs** to make agents topology-aware, achieving a **47% improvement** over baseline metrics using **Proximal Policy Optimization (PPO)**

### Exploring Sepsis detection using Diffusion Models | [GitHub](#)

- Engineered a **Conditional Denoising Diffusion Probabilistic Model (DDPM)** for medical time-series, treating vital sign forecasting as a generative "re-painting" task to visualize outcome distributions.
- Implemented a 1D ResNet-style architecture in PyTorch to process temporal dependencies and encode diffusion time-steps via sinusoidal embeddings
- Demonstrated **superior clinical utility** over traditional LSTMs by providing a "cloud" of potential patient outcomes.

**Deafine - Real-time Speaker Detection** | [GitHub](#)

- A simple Python accessibility tool that provides real-time speaker detection and live transcription using **ElevenLabs Speech-To-Text API** for deaf and hard-of-hearing users.
- Implemented realtime audio processing with **WebSocket** and made sure that there were no redundant API calls when there is silence through batched API calls.
- Secured the Best Use of ElevenLabs Award at **MLH-UB Hackathon**

**Multi Modal Genre Prediction** | [GitHub](#)

- Engineered a fusion model integrating textual plot summaries (via **LSTM**) and visual movie posters (via **MLP**) to enhance classification accuracy on a dataset of **100,000+ movies**.
- Implemented custom data augmentation and undersampling techniques to solve class imbalance across six main genres, **resulting in a 72% accuracy rate** for the integrated **multi-modal system**.

**LEADERSHIP AND ACTIVITIES**

**Light De Literacy NGO**  
*Admissions Head*

2020 - 2024

- Served as Admissions Head, facilitating the entry process for underprivileged children and coordinating large-scale fundraising and donation drives.

**Progno.AI Startup Initiative**  
*Team Lead*

2023

- Led a team to design a platform for early detection of lifestyle diseases, ensuring continuity of care and improved patient experiences.

**SKILLS**

- **Languages:** Python, SQL
- **Infrastructure:** Docker, Docker Compose, Kubernetes, REST APIs (FastAPI), Git, ETL, Spark
- **AI/Data:** PyTorch, TensorFlow, LLMs, Reinforcement Learning, Neural Networks, PostgreSQL, Pandas, NLP, Copilot, dbt, Generative AI, Diffusion Models, GNN, Computer Vision, RAG, Scikit Learn, LLM, Deep Learning
- **Analytical:** Statistical Modeling, Data Visualization (Plotly, Seaborn), Data Manipulation, Data Mining, Data Pipelines, Performance Tuning